

Hallucination Detection Report

AI Output Verification · Knowledge Graph Analysis

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Triples: 10 · Verdicts: 10 · Inconsistencies: 0 ·
Violations: 0

EXECUTIVE SUMMARY

1 Correct	6 Rule Violation	1 Unverifiable
1 Predicate Mismatch	1 Fabricated Fact	

EXTRACTED TRIPLES

All verifiable triples extracted from the LLM response before canonicalization or checking.

Subject	Predicate	Object	Verifiable	Verification Type	Predicate Definition
300	HAS_INTEREST_RATE	20	Yes	FACTUAL_CLAIM	Specifies the interest rate associated with the subject.
850	HAS_INTEREST_RATE	20	Yes	FACTUAL_CLAIM	Specifies the interest rate associated with the subject.
333	HAS_INTEREST_RATE	30	Yes	FACTUAL_CLAIM	Specifies the interest rate associated with the subject.
300	HAS_RISK_TIER	near prime	Yes	FACTUAL_CLAIM	Indicates the risk tier classification assigned to the subject.
850	HAS_RISK_TIER	prime	Yes	FACTUAL_CLAIM	Indicates the risk tier classification assigned to the subject.
305	HAS_RISK_TIER	subprime	Yes	FACTUAL_CLAIM	Indicates the risk tier classification assigned to the subject.
paris	IS_THE_CAPITAL_OF	france	Yes	FACTUAL_CLAIM	Indicates that the subject is the capital city of the object.
850	HAS_DOWN_PAYMENT	4	Yes	FACTUAL_CLAIM	Specifies the down payment amount associated with the subject.
850	HAS_DOWN_PAYMENT	2	Yes	FACTUAL_CLAIM	Specifies the down payment amount associated with the subject.
335	HAS_DOWN_PAYMENT	2	Yes	FACTUAL_CLAIM	Specifies the down payment amount associated with the subject.

CANONICALIZED TRIPLES

Predicates normalized against the database schema. Shows original predicate, canonical match, and match confidence.

Subject	Original Predicate	Object	Canonical Predicate	Confidence
300	HAS_INTEREST_RATE	20	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	HIGH
850	HAS_INTEREST_RATE	20	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	HIGH
333	HAS_INTEREST_RATE	30	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	HIGH
300	HAS_RISK_TIER	near prime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	HIGH
850	HAS_RISK_TIER	prime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	HIGH

Subject	Original Predicate	Object	Canonical Predicate	Confidence
305	HAS_RISK_TIER	subprime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	HIGH
paris	IS_THE_CAPITAL_OF	france	IS_THE_CAPITAL_OF	LOW
850	HAS_DOWN_PAYMENT	4	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	HIGH
850	HAS_DOWN_PAYMENT	2	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	HIGH
335	HAS_DOWN_PAYMENT	2	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	HIGH

LOGICAL INCONSISTENCIES

Contradictions detected between triples within the LLM response, independent of the database.

No logical inconsistencies detected.

SYMBOLIC RULE VIOLATIONS

Violations caught by explicit domain constraints before database lookup. Violated triples were removed from the evidence retrieval step.

No symbolic rule violations detected.

DETECTED HALLUCINATIONS

Subject	Predicate	Object	Verdict	Match	Violated	Conf.	Explanation
{ "subject": "300", "predicate": "INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE", "object": "20" }			Rule Violation	STRICT	OBJECT	1.00	The object '20' does not match the DB candidate object '22.94'. Evidence: { "subject": "300", "predicate": "INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE", "object": "22.94" }
{ "subject": "850", "predicate": "INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE", "object": "20" }			Rule Violation	STRICT	OBJECT	1.00	The object '20' does not match the DB candidate object '4.36'. Evidence: { "subject": "850", "predicate": "INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE", "object": "4.36" }
{ "subject": "333", "predicate": "INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE", "object": "30" }			Fabricated Fact	NONE	—	0.00	No DB candidates found for the given triple.
{ "subject": "300", "predicate": "RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL", "object": "near prime" }			Rule Violation	SEMIC	OBJECT	1.00	The object 'near prime' does not match the DB candidate object 'deep subprime'. Evidence: { "subject": "300", "predicate": "RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL", "object": "deep subprime" }

Subject	Predicate	Object	Verdict	Match	Violated	Conf.	Explanation
{ "subject": "850", "predicate": "RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL", "object": "prime" }			Rule Violation	SEM	OBJECT	1.00	The object 'prime' does not match the DB candidate object 'super prime'. Evidence: { "subject": "850", "predicate": "RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL", "object": "super prime" }
				ANTI			
				C			
{ "subject": "305", "predicate": "RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL", "object": "subprime" }			Rule Violation	SEM	OBJECT	1.00	The object 'subprime' does not match the DB candidate object 'deep subprime'. Evidence: { "subject": "305", "predicate": "RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL", "object": "deep subprime" }
				ANTI			
				C			
{ "subject": "paris", "predicate": "IS_THE_CAPITAL_OF", "object": "france" }			Unverifiable	NON	—	0.00	No DB candidates found and the predicate is not known.
				E			
{ "subject": "850", "predicate": "DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION", "object": "4" }			Rule Violation	STRICT	OBJECT	1.00	The object '4' does not match the DB candidate object '2.0'. Evidence: { "subject": "850", "predicate": "DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION", "object": "2.0" }

Subject	Predicate	Object	Verdict	Match	Violated	Conf.	Explanation
{ "subject": "850", "predicate": "DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION", "object": "2" }			Correct	SEMANTIC	—	1.00	The object '2' semantically matches the DB candidate object '2.0'. Evidence: { "subject": "850", "predicate": "DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION", "object": "2.0" }
{ "subject": "335", "predicate": "DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION", "object": "2" }			Predicate Mismatch	NON EXIST	PREDICATE	0.60	The predicate 'DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION' does not exist for the subject '335' in the DB candidates. Evidence: { "subject": "335", "predicate": "RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL", "object": "deep subprime" }

TRIPLE AUDIT TRAIL — ORIGINAL → CANONICAL → VERDICT

Full pipeline trace for every triple. UNVERIFIABLE* means the triple had no verdict — e.g. dropped due to output truncation.

Orig. Subject	Orig. Predicate	Orig. Object	Confidence	Canon. Predicate	Canon. Object	Verdict
300	HAS_INTEREST_RATE	20	HIGH	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	20	UNVERIFIABLE*
850	HAS_INTEREST_RATE	20	HIGH	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	20	UNVERIFIABLE*
333	HAS_INTEREST_RATE	30	HIGH	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	30	UNVERIFIABLE*
300	HAS_RISK_TIER	near prime	HIGH	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	near prime	UNVERIFIABLE*
850	HAS_RISK_TIER	prime	HIGH	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	prime	UNVERIFIABLE*
305	HAS_RISK_TIER	subprime	HIGH	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	subprime	UNVERIFIABLE*
paris	IS_THE_CAPITAL_OF	france	LOW	IS_THE_CAPITAL_OF	france	UNVERIFIABLE*
850	HAS_DOWN_PAYMENT	4	HIGH	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	4	UNVERIFIABLE*
850	HAS_DOWN_PAYMENT	2	HIGH	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	2	UNVERIFIABLE*

Orig. Subject	Orig. Predicate	Orig. Object	Confidence	Canon. Predicate	Canon. Object	Verdict
335	HAS_DOWN_PAYMENT	2	HIGH	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	2	UNVERIFIABLE*

VERDICT LEGEND

Verdict	Description
Correct	All three components (subject, predicate, object) verified against DB.
Rule Violation	One or more components contradict DB evidence.
Multi-Rule Violation	Violation spans multiple DB triples simultaneously.
Unverifiable	No usable DB evidence found for any component.
Predicate Mismatch	Subject real, but stated predicate is not a valid relation for it.
Fabricated Fact	Triple asserts a relationship that does not exist anywhere in DB.